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Hippocampal integration and separation processes with different temporal and spatial dynamics during learning for associative memory

Minjae Kwon¹ | Sang Wan Lee^{1,2}  | Sue-Hyun Lee³ 

¹Department of Bio and Brain Engineering, College of Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea

²Program of Brain and Cognitive Engineering, College of Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea

³Department of Psychology, College of Social Sciences, Seoul National University, Seoul, Republic of Korea

Correspondence

Sue-Hyun Lee, Department of Psychology, College of Social Sciences, Seoul National University, Seoul 08826, Republic of Korea.
Email: suehyunlee25@gmail.com

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Abstract

The hippocampus is known to be critically involved in associative memory formation. However, the role of the hippocampus during the learning of associative memory is still controversial; while the hippocampus is considered to play a critical role in the integration of related stimuli, numerous studies also suggest a role of the hippocampus in the separation of different memory traces for rapid learning. Here, we employed an associative learning paradigm consisting of repeated learning cycles. By tracking the changes in the hippocampal representations of associated stimuli on a cycle-by-cycle basis as learning progressed, we show that both integration and separation processes occur in the hippocampus with different temporal dynamics. We found that the degree of shared representations for associated stimuli decreased significantly during the early phase of learning, whereas it increased during the later phase of learning. Remarkably, these dynamic temporal changes were observed only for stimulus pairs remembered 1 day or 4 weeks after learning, but not for forgotten pairs. Further, the integration process during learning was prominent in the anterior hippocampus, while the separation process was obvious in the posterior hippocampus. These results demonstrate temporally and spatially dynamic hippocampal processing during learning that can lead to the maintenance of associative memory.

KEYWORDS

fMRI, hippocampus, integration process, learning, long-term memory, pattern separation

1 | INTRODUCTION

The process of associating related information is essential for memory formation. In explicit memory formation in particular, the hippocampus is known to play a crucial role in the association process (Alvarez & Squire, 1994; Corkin, 2002; Diana et al., 2007; Eichenbaum, 2004; Henke et al., 1999; Qin et al., 2009; Wirth et al., 2003). Prior neuroimaging studies have shown that hippocampal activity during encoding or immediately after encoding is correlated with

subsequent retrieval performance (Ben-Yakov & Dudai, 2011; Davachi, 2006; Jackson & Schacter, 2004; Kim, 2011; Kuhl et al., 2012; Sperling et al., 2003; Watanabe et al., 2011). Moreover, recent studies based on the multivoxel pattern analysis (MVPA) suggest a change in hippocampal representations after learning (Aly & Turk-Browne, 2016; Schlichting et al., 2015; Sun et al., 2021; Theves et al., 2019). A number of such studies have reported that pattern similarity between related stimuli in the hippocampus increased after learning and that the increased pattern similarity was associated with

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subsequent memory performance (Cooper & Ritchey, 2019; Schlichting et al., 2014; Wolosin et al., 2013; Zeithamova et al., 2018). Such findings therefore suggest that neural processing in the hippocampus is critical for the integration of distributed stimulus information (Backus et al., 2016; Gutchess & Schacter, 2012; Horner et al., 2015; Kim et al., 2017; Libby et al., 2019; Thavabalasingam et al., 2018; Wagner et al., 2020). However, the opposite tendency regarding changes of hippocampal representations has also been reported (Bakker et al., 2008; Kyle et al., 2015; Stokes et al., 2015; Yassa & Stark, 2011). A substantial number of studies have reported a learning-induced decrease in hippocampal pattern similarity of related stimuli, suggesting that greater learning-related pattern differentiation between related stimuli in the hippocampus was associated with better behavioral memory performance later (Hulbert & Norman, 2015; Jenkins & Ranganath, 2016; LaRocque et al., 2013; Wing et al., 2020). These studies suggest that the pattern separation process in the hippocampus may contribute to rapid memory formation by reducing interference between stimuli while the prefrontal cortex subserves a role in the integration process (Berens et al., 2018; Favila et al., 2016; Kirwan & Stark, 2007; Lohnas et al., 2018; O'Reilly et al., 2014).

To learn new associative information successfully, both the extraction of general properties and distinguishing different features for associated stimuli should be important. Thus, it is plausible that both integration and separation processes may be involved during learning even for the same task. However, while prior studies suggest a role of the hippocampus in the integration (Hsieh et al., 2014; Wagner et al., 2020; Wolosin et al., 2013; Zeithamova et al., 2018) or separation (Berens et al., 2018; Favila et al., 2016; Jenkins & Ranganath, 2016; LaRocque et al., 2013) process during learning, it remains unclear as to whether both integration and separation processes can occur during the same learning procedure in the hippocampus. Prior studies compared the hippocampal representations before and after learning, but did not examine the temporal changes of the hippocampal representation during learning (Favila et al., 2016; Schapiro et al., 2012; Schlichting et al., 2015). If integration or separation processes occur temporally during learning, such temporal changes may not be detected in comparisons of only before and after learning without tracking intermediate processes. One recent potent proposal for hippocampal roles during learning suggested that hippocampal representations for related memories are flexible, depending on behavioral goals (Brunec et al., 2020). It was suggested that the similarity in neural representations of related stimuli increases when the current goal is to extract shared information, whereas the degree of representational similarity decreases when the goal is to separate related stimuli. Although this suggestion mainly focuses on the goal-dependent hippocampal processing that occurs unidirectionally during a task, it may be possible that such goal-dependent processing in the hippocampus not only occurs in one direction during the same task, but also varies from moment to moment depending on certain temporal goals. Hence, here we tested the possibility that both integration and separation processes occur temporally during learning in the hippocampus, contributing to the strengthening of memory.

To investigate changes in hippocampal representations for associative stimuli during learning, we tracked temporal changes at the neural pattern similarity level between paired stimuli in the hippocampus as learning progressed. To do this, we performed an fMRI study combined with an associative memory task, which consisted of a Learning session and two subsequent Retrieval sessions (Figure 1a). During the Learning session, participants were asked to learn object-building image pairs while their brain activity was measured by fMRI. The Learning session consisted of four repeated cycles, which allowed us to track the temporal changes in the hippocampal representation of each stimulus as learning progressed (Figure 1a). To measure temporal changes in neural representations during learning, we used a trial-by-trial similarity analysis (Visser et al., 2013; Figure 2a) based on the MVPA (Kriegeskorte et al., 2008; Figure 2b) and tracked the changes in neural pattern similarity levels between paired stimuli across the repeated learning cycles (Figure 1a,b). One day and four weeks after the scan, the participants completed the Retrieval sessions, in which their retrieval performance for the learned associations was measured (Figure S1a). In this study, we show both increases and decreases in pattern similarity between paired stimuli during learning in the hippocampus, indicating the existence of both integration and separation processing in the hippocampus during learning. Moreover, subsequent memory performance 1 day or 4 weeks after learning depends on this dynamic change of hippocampal representations during learning. Furthermore, consistent with the prior research that suggests differential roles of the anterior and posterior hippocampus (Brunec et al., 2018; Poppenk et al., 2013; Robin & Moscovitch, 2017; Schlichting et al., 2015), we also show the different involvement of the anterior and posterior hippocampus in integration and separation processes.

2 | METHODS

2.1 | Participants

Thirty-one healthy and right-handed participants (mean age = 22.71 ± 0.26 years, 14 females) took part in the fMRI experiment. Inclusion criteria included normal or corrected-to-normal vision, a minimum of high school graduation, and no history of neurological or psychiatric disorders. Additionally, all participants were native speakers of the Korean language. They all provided written informed consent for the procedure in accordance with protocols approved by the Korea Advanced Institution of Science and Technology (KAIST) Institutional Review Board. In the comparison between the remembered and forgotten pairs, the data from 22 of the participants were used because the other participants remembered or forgot all pairs in the 1-day Retrieval or 4-week Retrieval session. The exclusion of these participants was based solely on their behavioral performance without any prior knowledge about each individual.

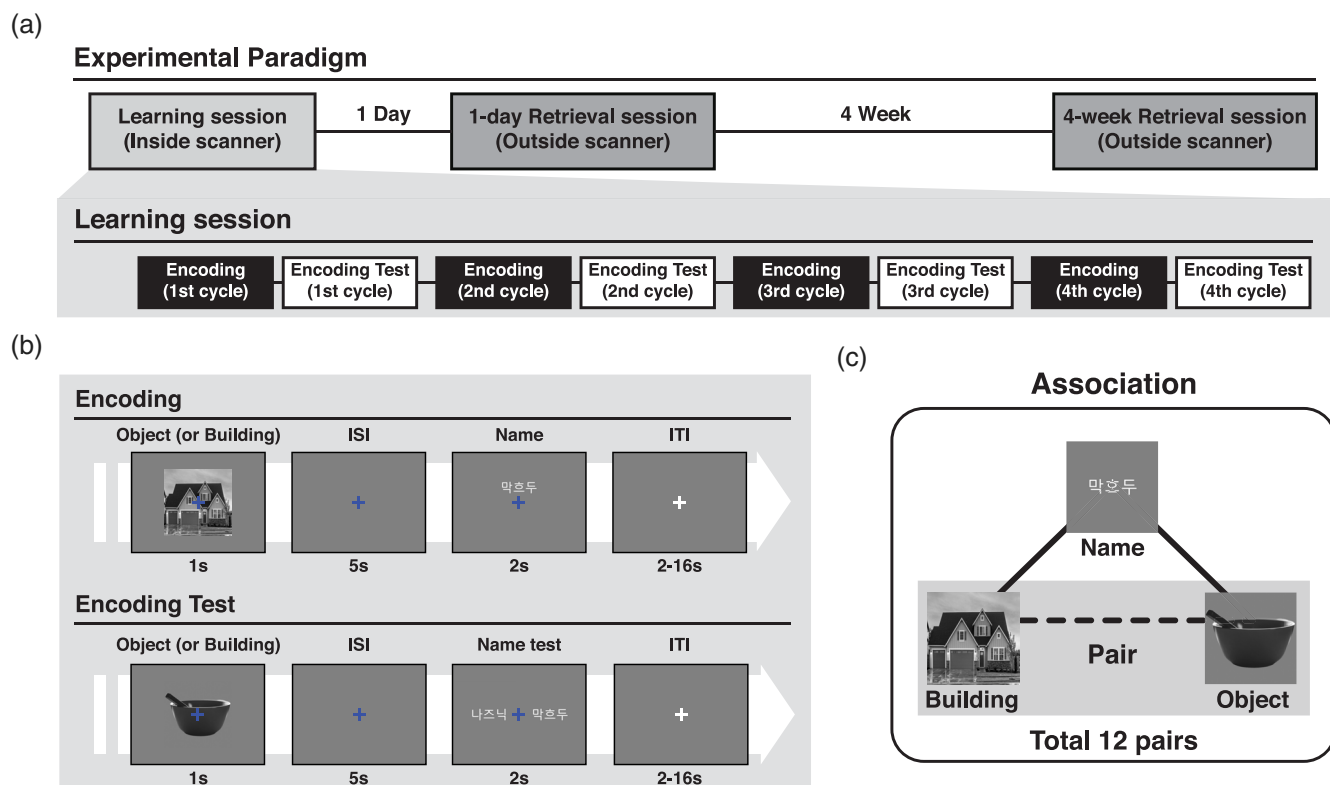


FIGURE 1 Experimental procedures. (a) Participants completed an associative memory task, which consisted of a Learning session and subsequent two Retrieval sessions, during which their retrieval performance was measured 1 day (1-day Retrieval session) and 4 weeks (4-week Retrieval session) after the Learning session. Only the Learning session was conducted inside the MRI scanner. In the Learning session, “Encoding” and the following “Encoding Test” runs were repeated four times. (b) During each Encoding trial, the participants viewed one of the 12 objects or 12 buildings, followed by the presentation of an associated name. They were asked to memorize 12 name-object-building associations. In each trial of the Encoding Test runs, the participants viewed one of the 12 objects or 12 buildings, followed by the presentation of two names. They were instructed to choose the name associated with the presented object or building. (c) For each association, a building and an object were associated with a common name. During the Learning session, the participants eventually learned 12 fixed object-building pairs.

2.2 | Task and stimuli

Participants completed an associative memory task consisting Learning, 1-day Retrieval, and 4-week Retrieval sessions. The Learning session was conducted inside an MRI scanner, while the 1-day Retrieval and 4-week Retrieval sessions were conducted in a behavioral testing room outside the scanner 1 day and 4 weeks after the Learning session, respectively (Figure 1a).

2.2.1 | Learning session

During the Learning session, “Encoding” and the following “Encoding Test” runs were repeated four times (Figure 1a). During the Encoding runs, the participants were instructed to memorize 12 name-object-building associations.

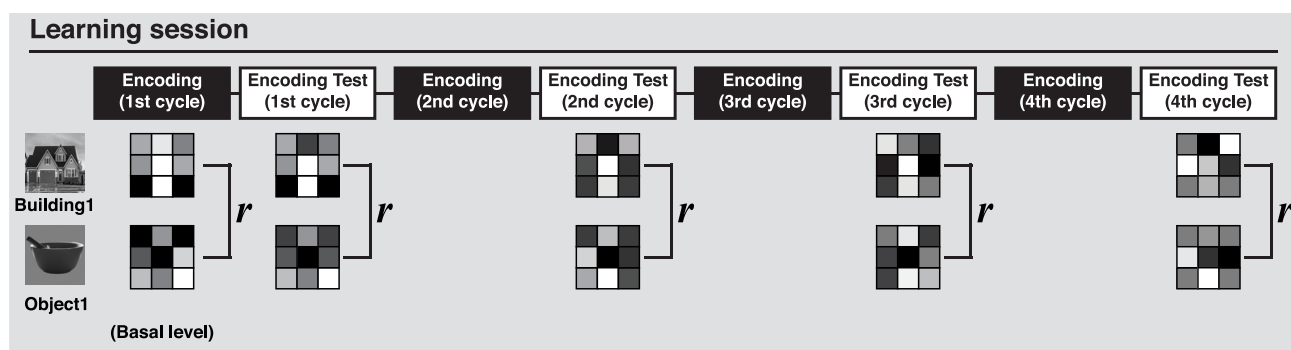
Each run of Encoding consisted of 24 trials, comprising 12 trials for object-name associations and 12 trials for building-name associations, resulting in one trial for each association per run and four repetitions per association across all four runs. Through this Encoding design, which entailed associating one name with both an object and

a building (Figure 1c), the participants eventually learned 12 fixed object-building pairs during the Encoding runs. The order of the associations was randomized and counterbalanced across runs. During each trial of Encoding run, the participants first viewed one of the 12 objects or 12 buildings for 1 s. Then, after an interstimulus interval of 5 s, the associated name was presented for 2 s, followed by a jittered intertrial interval (ITI) ranging from 2 to 16 s (average ITI = 8 s) (Figure 1b).

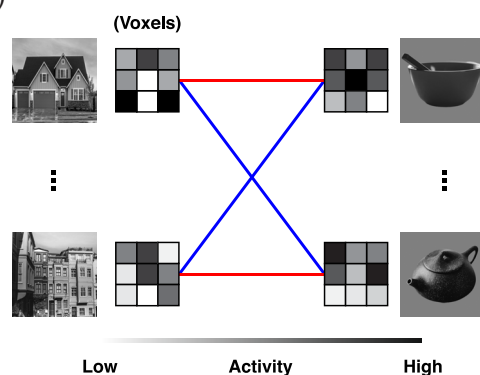
The design of the Encoding Test run was identical to that of the Encoding run, with the exception that two different names were presented on the left and right sides of the screen, instead of presentation of a single name. Participants were instructed to choose the name associated with the previously shown image between the presented names (Figure 1b). During the Encoding Test runs, feedback on the participants' responses were not provided to prevent additional encoding. Each run of the Encoding or Encoding Test had a duration of 378 s (189 TRs), and the entire Learning session, which consisted of four Encoding-Encoding Test cycles, took a total of 3024 s.

The name stimuli were pseudo-names that we generated by modifying selected African-region names, which were unfamiliar to Korean participants. The pseudo-names were presented using the Korean

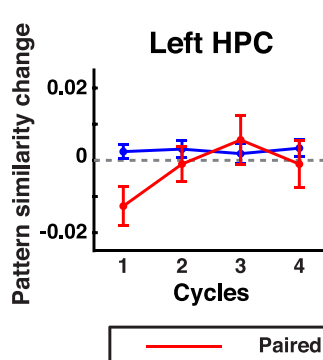
(a)



(b)



(c)



(d)

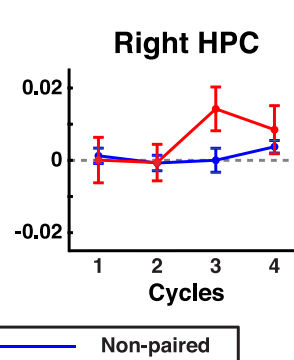


FIGURE 2 Pattern similarity changes during the Learning session between paired and non-paired stimuli in the hippocampus (HPC). (a) The neural response patterns for each object or building during the Encoding Test were used for a trial-by-trial similarity analysis. The pattern similarity based on the Pearson correlation coefficients (r) was derived from the response patterns of object-building pairs within each run of the Encoding Test. The change in the degree of pattern similarity (pattern similarity change) for each pair from the basal level was calculated at each Encoding Test cycle. (b) Calculations of the degree of pattern similarity for paired (red line) and non-paired (blue line) object-building stimuli. (c, d) Pattern similarity changes for paired (red line) and non-paired (blue line) object-building stimuli during each cycle of the Encoding test in the left HPC (c) and the right HPC (d). There was a significant interaction in a two-way ANOVA between cycle (1–4) and pairing type (paired, non-paired) for the left ($p = .046$) and right HPC ($p = .050$).

alphabet (Hangul). We used 12 images of kitchen tools and 12 images of buildings (Figure 1c). These images were chosen because they were considered to differ sufficiently in terms of their orientation, subcategory, and shape, and to avoid stereotypical shapes, so that there would be minimal effects of prior knowledge or interference between them. The image size was 500×500 pixels, and each image was presented at a visual angle of 7° with a uniform gray background.

Throughout the Encoding runs or Encoding test runs, the participants were asked to remain fixated on a central cross. The color of the fixation cross was blue throughout each trial and white during each ITI.

2.2.2 | Retrieval session

The 1-day Retrieval session and 4-week Retrieval session were identical except for the duration between the Learning session and each Retrieval session; 1-day Retrieval and 4-week Retrieval sessions were conducted 1 day and 4 weeks after the Learning session, respectively. During each Retrieval session, the participants mainly conducted a

Pair recognition test, where they viewed one object and one building that were simultaneously presented, and had to answer whether the presented object and building were associated with the same name by pressing a yes or no button within 10 s (Figure S1a). There was a total of 24 questions; the answer to 12 questions was yes (one question for each associated object-building pair), and the answer to the remaining 12 questions was no (questions with 12 nonassociated pairs). All of the images used in this test were the object or building images that the participants learned in the Learning session. Each image was presented two times throughout the test, and the order of the associated or nonassociated pairs was randomized; for each object-building association, there were three related questions (one “yes” question and two “no” questions). After responding to each question, the participants were asked to choose their subjective confidence level for their answers from the choices of 100, 70, 40, and 10% within 5 s, followed by an interval of 5 s. When the score for each question is considered as being independent, the average accuracy rate for the Pair recognition test conducted during the 1-day Retrieval session was $82.8 \pm 3.35\%$, while this rate was $72.3 \pm 3.11\%$ for the test conducted in the 4-week Retrieval session, showing

significantly higher levels than the chance level (50%) (for the 1-day Retrieval session, two-tailed t test, $t_{(30)} = 9.937$, $p < .001$; for the 4-week Retrieval session, two-tailed t test, $t_{(30)} = 7.295$, $p < .001$). However, because the three questions were related to each object-building association, we calculated the accuracy based on the consideration that each object-building association is remembered only when the answers to all these three questions were correct (Figure S1a). Based on this method, the average accuracy rate for the Pair recognition test conducted during the 1-day Retrieval session was $67.5 \pm 5.86\%$, while it was $48.7 \pm 5.03\%$ for the test conducted during the 4-week Retrieval session (Figure S1b). These retrieval performances were also significantly greater than the chance level (12.5%) (for the 1-day Retrieval session, two-tailed t test, $t_{(30)} = 9.540$, $p < .001$; for the 4-week Retrieval session, two-tailed t test, $t_{(30)} = 7.304$, $p < .001$). This method of calculating the accuracy is applied to all of the following analyses. Before the Pair recognition test, the participants were asked to recall the associated name for each given (object or building) image. The participants remembered $44.89 \pm 5.36\%$ of the associated names accurately during the 1-day Retrieval session and $19.89 \pm 3.08\%$ of the associated names during the 4-week Retrieval session. They remembered two of the three syllables for $54.44 \pm 5.66\%$ of the names during the 1-day Retrieval session and $24.87 \pm 3.73\%$ during the 4-week Retrieval session. After the Pair recognition test, they completed a test identical to the Encoding Test conducted inside the scanner. They showed $93.41 \pm 1.99\%$ accuracy on this test during the 1-day Retrieval session and $87.37 \pm 2.29\%$ accuracy on the same test during the 4-week Retrieval session.

2.3 | fMRI data acquisition

MRI measurements were obtained with a 3 T Siemens Prisma scanner at the Center for Neuroscience Imaging Research of the Institute for Basic Science. Whole brain T2*-weighted echo-planar imaging data were acquired using a 64-channel head coil with an in-plane resolution of 2.5×2.5 mm and 48 4.5 mm slices: 0.25 mm interslice gap, a field of view (FOV) = 192 mm, flip angle = 90° , repetition time (TR) = 2 s, echo time (TE) = 17 ms. After experimental runs, standard magnetization-prepared rapid-acquisition gradient echo images were collected with an in-plane resolution of 1×1 mm (1-mm slices, FOV = 256 mm, flip angle = 8° , TR = 2 s, TE = 2.45 ms) to get a high-resolution anatomical data.

2.4 | fMRI data analyses

2.4.1 | Preprocessing

The MRI data analysis was conducted using AFNI (<http://afni.nimh.nih.gov>) (Cox, 1996), SUMA (AFNI surface mapper), FreeSurfer, and custom MATLAB scripts. Data preprocessing included slice time

correction and motion correction. Then, the percent signal change was calculated for each run of the fMRI scan and each participant on a voxel-by-voxel basis by dividing the BOLD signal by the mean BOLD across the entire time points for each voxel.

2.4.2 | Trial-by-trial pattern similarity

To deconvolve the event-related BOLD responses during Encoding Test scan runs, we conducted a standard general linear model using the AFNI software package, “3dDeconvolve” with GAM function. The t -value of each voxel was derived from the onset of each object or building presentation.

To track how the neural patterns change as repetitive associative learning progressed, we employed trial-by-trial similarity analysis (Visser et al., 2013) based on MVPA (Kriegeskorte et al., 2008). In this analysis, the trials of the Encoding test runs were modeled as separate events, and the estimated t -values were used. The Pearson correlation coefficients (r) between all possible patterns of objects and buildings (both associated object-building pairs (paired) and nonassociated pairs (non-paired)) within each run were calculated. Correlations were then Fisher-transformed. The pattern similarity change of each pair at each trial was calculated by subtracting the correlation coefficient that was derived from the responses of the first exposure (basal level, responses at the trial of the first Encoding run) from the correlation coefficient derived from each following trial of Encoding test runs for each object-building pair.

2.4.3 | Regions of interest

The hippocampus was automatically defined by the subcortical parcellation of FreeSurfer. The segmentation of the hippocampus into anterior and posterior regions was performed based on the predefined cutoffs allocating anterior 35% of slices in the anterior-posterior direction to the anterior hippocampus (Hackert et al., 2002; Figure 4c). The rest of the slices was defined as the posterior. To assess the pattern similarity changes along the anterior-posterior axis, we defined subregions, with each subregion corresponding to 40% of the total volume of the hippocampus, and incrementally shifting the center of the subregion from the anterior to the posterior of the hippocampus. The subregions were centered at every 5% of the hippocampal voxels along the anterior-posterior axis.

2.5 | Statistical analysis

Repeated-measures ANOVAs (tests of within-subject effects) were used to determine the statistical significance of any cycle or condition effect. For all of the ANOVAs, the Greenhouse-Geisser correlation was used if the assumptions of sphericity were not met. Otherwise, the Sphericity Assumed correlation was used. To compare the pattern

similarity between remembered and forgotten pairs, two-tailed paired *t*-tests were conducted with correction for false discovery rates (FDRs) using the Benjamini–Hochberg procedure. To examine the contribution of different remembered and forgotten pair numbers to the degree of pattern similarity, permutation tests were conducted. During these tests, we randomly shuffled the pair labels (remembered vs. forgotten) while maintaining the number of remembered and forgotten pairs for each participant, and derived the outcomes from the neural responses induced by the randomized pairs. We repeated this process 100,000 times and tested whether the actual pattern similarity difference between the remembered and forgotten pairs fell within the top 5% of the simulated null distribution of pattern similarity differences.

3 | RESULTS

3.1 | Dynamic temporal changes of hippocampal representations during learning

To investigate how hippocampal representations of associative information change during learning, we examined changes in neural pattern similarity levels across repeated Encoding–Encoding Test cycles of the Learning session (Figures 1a and 2a). To track the change in memory-specific neural representations without interference from neural activity generated by the perception of paired stimuli or control processes such as monitoring and organization during Encoding, we primarily focused on the responses during Encoding Test runs, which were conducted immediately after each Encoding run. The neural response pattern measured during each Encoding Test following each Encoding may reflect the memory representation altered by the previous Encoding. Using a trial-by-trial pattern similarity analysis (Visser et al., 2013), we calculated the change in the degree of neural pattern similarity for each pair of stimuli on each Encoding Test cycle from the basal level (the pattern similarity derived from the first exposure to stimuli on the first run of Encoding) (Figure 2a). This subtraction of the basal level served to exclude any effects of internal representation arising from prior knowledge or preexisting associative memories of the participants. The pattern similarity changes of paired stimuli were then compared to those of non-paired stimuli (Figure 2b).

While the pattern similarity of non-paired stimuli was comparable to the basal level across the four cycles of the Encoding Test, the pattern similarity of paired stimuli fluctuated dynamically as learning progressed (one-way ANOVA across cycles (1–4), in the left hippocampus, $F_{(3,90)} = 2.976$, $p = .036$ for the paired stimuli, $F_{(3,90)} = 0.148$, $p = .931$ for the non-paired stimuli; in the right hippocampus, $F_{(3,90)} = 2.599$, $p = .057$ for the paired stimuli, $F_{(3,90)} = 1.195$, $p = .316$ for the non-paired stimuli). Moreover, we found a significant interaction in a two-way ANOVA between cycle (1–4) and pairing type (paired, non-paired) ($F_{(3,90)} = 2.765$, $p = .046$ for left hippocampus; $F_{(3,90)} = 2.710$, $p = .050$ for right hippocampus), indicating the different temporal profiles of pattern similarity between paired stimuli and non-paired stimuli (Figure 2c,d). These results

suggest that the hippocampal representations changed dynamically during the learning process.

3.2 | Hippocampal information processing during learning between remembered and forgotten pairs

The question arises as to whether the dynamic change of the hippocampal representations of paired stimuli during learning is related to the maintenance of associative memory. To answer this question, we next compared the pattern similarities across the Encoding Test cycles between remembered pairs and forgotten pairs at the time of the Retrieval sessions (Figures 1a, 3a, and S1). Notably, we found that the pattern similarity of the remembered pairs dynamically changed across the Encoding Test cycles, whereas little change was observed in the pattern similarity of forgotten pairs, as in the pattern similarity of non-paired stimuli (Figure 3). A two-way ANOVA with cycle (1–4) and the 1-day retrieval performance (remembered, forgotten) as factors revealed a significant interaction between cycle and retrieval performance ($F_{(3,63)} = 2.879$, $p = .043$ for right hippocampus; Figure 3b, right).

In the comparison between the remembered and forgotten pairs during the 1-day Retrieval session at each cycle, we found a significant difference between the outcomes on the third cycle of the Encoding Test in the right hippocampus: the pattern similarity of the remembered pairs was significantly greater than the pattern similarity of the forgotten pairs upon the third cycle ($t_{(21)} = 3.822$, FDR-corrected $p = .004$; Figure 3b). Moreover, this pattern similarity of remembered pairs on the third cycle was significantly greater than the basal level ($t_{(21)} = 3.453$, FDR-corrected $p = .010$; Figure 3b), whereas that of the forgotten pairs was comparable to the basal level ($t_{(21)} = -1.781$, FDR-corrected $p = .358$; Figure 3b).

We also compared the degrees of pattern similarity between remembered and forgotten pairs according to the 4-week retrieval performance. While the pattern similarity of remembered pairs based on the 1-day retrieval performance was significantly greater than the basal level on the third cycle of the Encoding Test (Figure 3b), the pattern similarity of remembered pairs, as determined by the 4-week retrieval performance, showed a tendency to increase from the basal level on the fourth cycle of the Encoding Test in the right hippocampus ($t_{(21)} = 2.423$, uncorrected $p = .025$, FDR-corrected $p = .098$; Figure 3c).

Because the numbers of remembered and forgotten pairs were different between the participants, we additionally examined whether these different pair numbers might contribute to the similarity of the patterns by randomizing the pair labels (remembered vs. forgotten) and computing the degree of pattern similarity from them. By repeating this approach, we confirmed that the main effect of pattern similarity stems from the different responses between remembered and forgotten pairs but not from the different pair numbers (Figure S2). In addition, we did not find any significant difference in the accuracy levels between the remembered and forgotten pairs on the last (fourth) cycle of the Encoding Test of the Learning session, as well as on other cycles, indicating comparable learning abilities between remembered and forgotten pairs (Figure S3d,e).

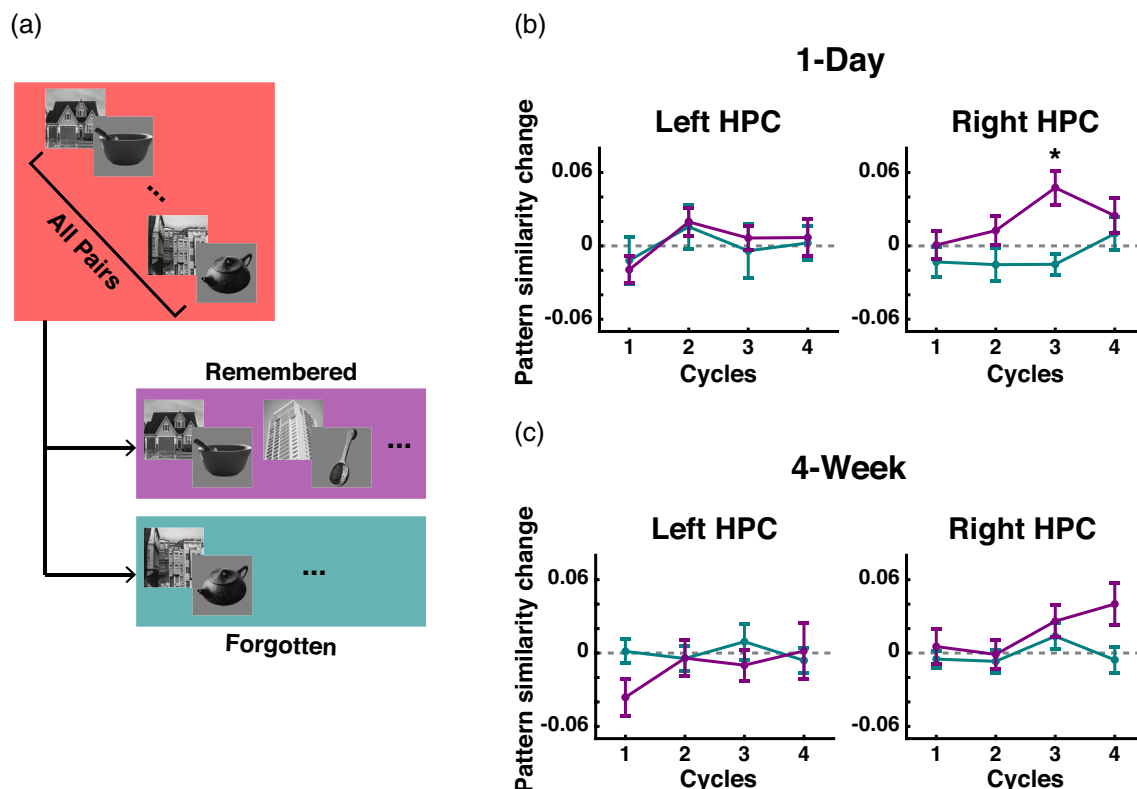


FIGURE 3 Pattern similarity changes during the Learning session between remembered and forgotten pairs in the hippocampus (HPC). (a) Pattern similarities across the Encoding Test cycles of the Learning session were compared between remembered pairs and forgotten pairs at the time of the 1-day and 4-week Retrieval sessions. (b) Changes in the degree of pattern similarity for 1-day remembered (purple line) and forgotten (green line) pairs at each cycle of the Encoding Test in the left and right HPC. (c) Changes in the degree of pattern similarity for 4-week remembered (purple line) and forgotten (green line) pairs at each cycle of the Encoding Test in the left and right HPC. * $p < .05$ (FDR-corrected) for the difference between the remembered and forgotten pairs. The dotted line indicates the basal level (zero). Error bars indicate between-subjects SEM.

3.3 | Differential processing of the anterior and posterior hippocampus during learning

Regarding the role of the hippocampus in memory processes, it has been suggested that the anterior hippocampus plays a role in integration (Brunec et al., 2018; Deuker et al., 2016; Schlichting et al., 2015; Theves et al., 2019), whereas the posterior hippocampus mainly supports separation processing (Berens et al., 2018; Brunec et al., 2018; Milivojevic et al., 2015; Schlichting et al., 2015) based on several imaging studies and structure studies (Dalton et al., 2019; Poppenk et al., 2013). To test whether the temporal increase in pattern similarity we observed during learning is prominent in the anterior hippocampus while the decrease in pattern similarity is mainly observed in the posterior hippocampus, we also examined cycle-by-cycle pattern similarity for paired stimuli during learning in the anterior and posterior hippocampus regions separately (Figure 4a–c).

In the right anterior hippocampus, we found that the pattern similarity of the remembered pairs, as determined by the 1-day retrieval performance, was significantly greater than the pattern similarity of the forgotten pairs upon the third cycle ($t_{(21)} = 3.791$, FDR-corrected $p = .004$; Figure 4a) but not in the right posterior hippocampus (1-day Retrieval session, third cycle, $t_{(21)} = 1.209$, FDR-corrected $p = .725$;

Figure 4b). Moreover, this tendency was also observed in the pattern similarities between the remembered and forgotten pairs determined by the 4-week retrieval performance upon the fourth cycle (right anterior hippocampus, $t_{(21)} = 3.649$, FDR-corrected $p = .006$; right posterior hippocampus, $t_{(21)} = -0.157$, FDR-corrected $p = .877$; Figure 4a,b). Additionally, we found a significant increase in the degree of pattern similarity for remembered pairs from the basal level in the right anterior hippocampus (1-day Retrieval session, third cycle, $t_{(21)} = 3.017$, FDR-corrected $p = .026$; 4-week Retrieval session, fourth cycle, $t_{(21)} = 3.298$, FDR-corrected $p = .014$; Figure 4a). On the other hand, in the left posterior hippocampus, the pattern similarity of the remembered pairs, as determined by the 4-week retrieval performance, was significantly lower than the pattern similarity of the forgotten pairs upon the first cycle ($t_{(21)} = -2.933$, FDR-corrected $p = .032$; Figure 4b). These results showed that the temporal increase in pattern similarity during learning for the remembered pairs was prominent in the anterior hippocampus of the right hemisphere, whereas the temporal decrease in pattern similarity for the remembered pairs was noticeable in the posterior hippocampus of the left hemisphere.

Focusing on the cycles when a significant difference between the remembered and forgotten pairs was observed, we also examined

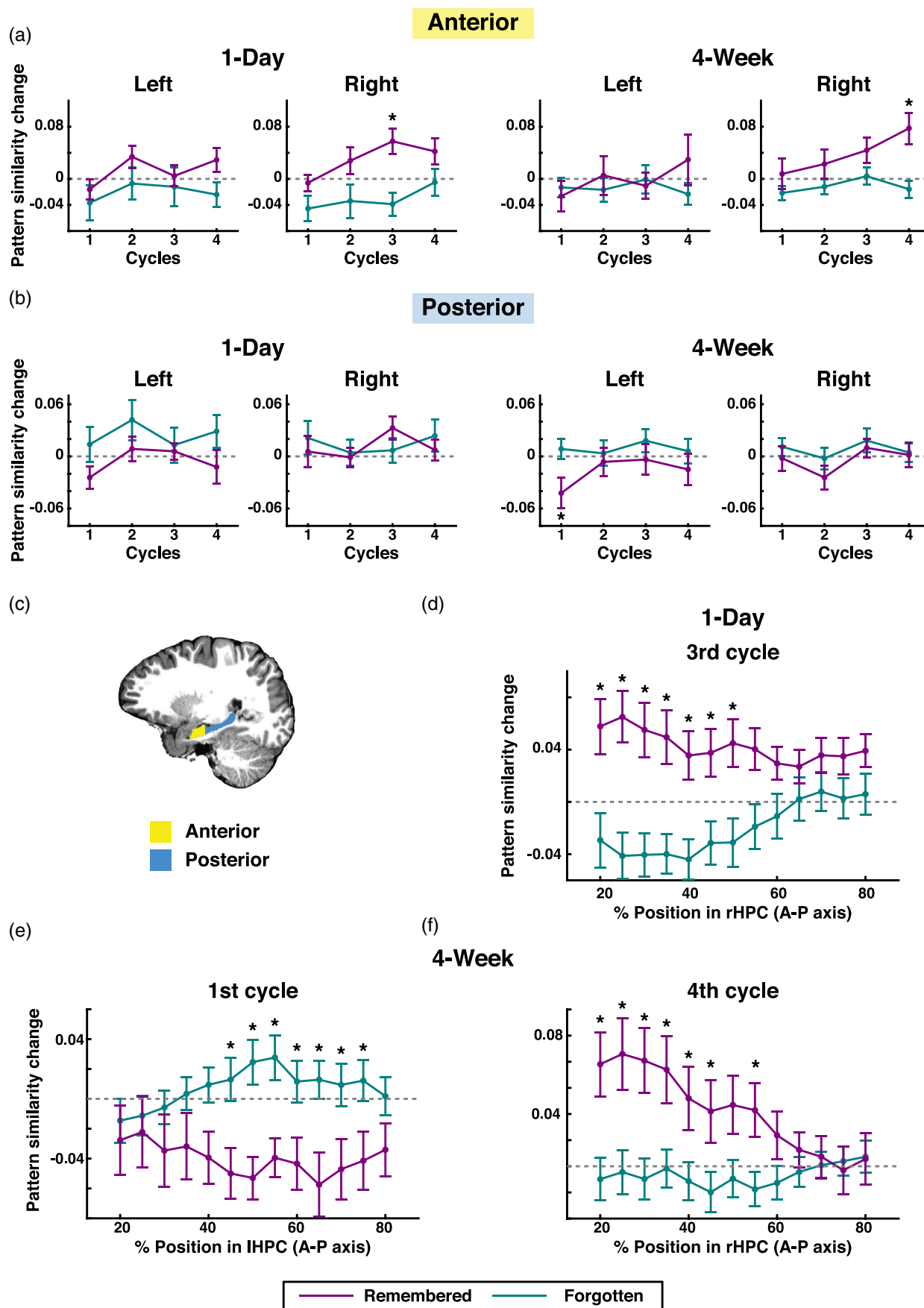


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pattern similarities at different locations of the hippocampus along its anterior–posterior axis (Figure 4d–f). Consistent with the results above, on the third cycle of the Encoding Test, the pattern similarity of the remembered pairs based on 1-day retrieval performance was greater than the pattern similarity of the forgotten pairs in the anterior and middle regions of the right hippocampus (all $t_{(21)} > 2.889$, FDR-corrected $p < .017$ for all positions between the anterior end and the position of 50% of voxels along the anterior–posterior axis; Figure 4d), while the differences between remembered and forgotten pairs were relatively weak in the posterior regions (Figure 4d). Moreover, based on the 4-week retrieval performance, higher pattern similarity for the remembered pairs compared to the forgotten pairs was also prominent in the anterior and middle regions of the right hippocampus on the fourth cycle of the Encoding Test (all $t_{(21)} > 2.407$, FDR-corrected $p < .048$ for all positions between the anterior end and the positions of 55% of voxels except for the position of 50%; Figure 4f), whereas lower pattern similarity of remembered pairs compared to forgotten pairs was noticeable in the middle and posterior regions of the left hippocampus on the first cycle of the Encoding Test (all $t_{(21)} < -2.644$, FDR-corrected $p < .029$ for all positions between 45 and 75% of voxels along the anterior–posterior axis; Figure 4e).

Taken together, these results show that temporal increases and decreases in pattern similarity for associative information during learning occur at different locations of the hippocampus. These findings suggest that the integration process is predominant in the right anterior hippocampus during the late phase of learning, while the differentiation process prevails in the left posterior hippocampus during the early phase of learning. Further, they also suggest that both the integration process of the anterior hippocampus and the differentiation process of the posterior hippocampus during learning are associated with subsequent memory retrieval.

4 | DISCUSSION

Our results show that hippocampal representations of associative information change dynamically during learning, involving both temporal increases and decreases in pattern similarity levels between associative stimuli. Moreover, these dynamic temporal changes in pattern similarity during learning were associated with subsequent memory

strength. Further, while the decrease in pattern similarity for associated pairs was obvious at the earlier phase of learning in the posterior hippocampus, the increase in pattern similarity was prominent at the later phase of learning in the anterior hippocampus. These findings suggest the existence of temporally and spatially different initial neural processes in the hippocampus, leading to the maintenance of memory.

Although previous research has suggested that hippocampal processing during learning is related to subsequent memory performance (Kuhl et al., 2012; Watanabe et al., 2011), the interpretation for the role of the hippocampus during learning was fragmentary. Several studies have argued that pattern separation during learning is related to memory performance (Favila et al., 2016; LaRocque et al., 2013; Wing et al., 2020), whereas other studies suggest that pattern integration during learning is critically associated with memory (Cooper and Ritchey, 2019; Schlichting et al., 2014; Wolosin et al., 2013; Zeithamova et al., 2018). Although pattern integration and pattern separation appear to be opposite types of information processing, both extraction of general properties and distinguishing different features should be essential for successful associative learning. However, because these studies compared neural responses before and after learning, whether both integration and separation processes are involved during learning remains unclear. If one or both processes are temporally involved during learning in the hippocampus, such transient changes may not be detectable without monitoring hippocampal responses as learning progresses. In the present study, we showed cycle-by-cycle changes in hippocampal representations of associative information during learning. Our results suggest the possibility that even during the learning of the same task, both of the pattern integration and the pattern separation associated with subsequent memory performance can occur at different time points during learning. At an earlier phase of learning, the pattern separation process seems to be prominent, while at a later phase, integration processing for forming associative memory may be noticeable.

To reconcile the different results of the hippocampal function during learning, one recent compelling proposal is that the role of the hippocampus depends on the current behavioral goal (Brunec et al., 2020). Our results are also consistent with this argument. In particular, for the pairs maintained until the 1-day or 4-week Retrieval session, we revealed that pattern similarity between the associated stimuli increased as learning progressed, resulting in significantly

FIGURE 4 Different pattern similarity changes during the Learning session between remembered and forgotten pairs in the anterior and posterior hippocampus (HPC). (a) Pattern similarity changes for each cycle of the Encoding Test in the anterior hippocampus (aHPC) for remembered (purple line) and forgotten (green line) pairs at the time of the 1-day Retrieval and 4-week Retrieval sessions. (b) Pattern similarity changes for each cycle of the Encoding Test in the posterior hippocampus (pHPC) for remembered (purple line) and forgotten (green line) pairs at the time of the 1-day Retrieval and 4-week Retrieval sessions. (c) The hippocampus, delineated on a participant's T1-weighted image. Yellow and blue areas indicate voxels included in the anterior and posterior parts of the hippocampus, respectively. (d) Pattern similarity changes for 1-day remembered (purple line) and forgotten (green line) pairs on the third cycle of the Encoding test at different locations of the right hippocampus (rHPC) along the anterior–posterior axis. (e) Pattern similarity changes for 4-week remembered (purple line) and forgotten (green line) pairs on the first cycle of the Encoding test at different locations of the left hippocampus (lHPC) along the anterior–posterior axis. (f) Pattern similarity changes for 4-week remembered (purple line) and forgotten (green line) pairs on the fourth cycle of the Encoding test at different locations of the rHPC along the anterior–posterior axis. * $p < .05$ (FDR-corrected) for the difference between the remembered and forgotten pairs. The dotted line indicates the basal level (zero). Error bars indicate between-subjects SEM.

greater pattern similarity than the degree of pattern similarity for forgotten pairs (or the basal level) on the third or fourth cycle of the Encoding Test (Figure 3). This increase in pattern similarity corresponds to the final goal of the task in which a participant needs to associate an object image and a building image with a common name. However, our results show that even during learning that eventually converges on pattern integration, a pattern separation process also occurs in the early phase of learning. Both pattern integration and pattern separation processes may exist in parallel in the hippocampus with different temporal profiles. Immediately after the first encoding cycle, the pattern separation process may be more active to prevent any interference between stimuli and to assess new information correctly. Finally, depending on the behavioral goal, net hippocampal processing may be biased toward one of the two processes.

In addition to the temporal dynamics of hippocampal functions, our results also suggest that different subfields of the hippocampus preferentially subserve different roles. We showed that the earlier separation of representations is prominent in the posterior hippocampus whereas the later integration of representations is obvious in the anterior hippocampus. These results are consistent with prior research showing that the anterior hippocampus is involved in pattern integration (Robin & Moscovitch, 2017; Schlichting et al., 2015; Theves et al., 2019) and that the activity of the posterior hippocampus is associated with pattern separation (Berens et al., 2018; Robin & Moscovitch, 2017; Schlichting et al., 2015). Our results extend these prior findings, showing that different parts of the hippocampus temporally subserve differential roles of information processing during learning. The lateralization of the hippocampal processing in our results may reflect the specific properties of the stimuli we used in the tasks or the participants' strategies for learning. Further research will be needed to clarify the neural processing underlying the lateralization of the hippocampus during associative learning.

Our results also showed that the degree of pattern similarity for the remembered pairs based on the performance at the 1-day Retrieval session was highest after the third encoding cycle, while that for the remembered pairs based on the 4-week Retrieval session gradually increased over cycles and was highest after the fourth encoding cycle. These results indicate that information processing in the hippocampus at different time points during learning may contribute differently to the strengthening of associative memory. Further, these results propose the possibility that we can predict how long the memory will last based on the hippocampal responses at the time of learning. In the analyses, we employed stringent criteria for remembered pairs to evaluate definite associative memory, resulting in the exclusion of nine participants who either remembered or forgot all pairs during the 1-day Retrieval or 4-week Retrieval session. In future studies, adjusting the task difficulty level might be necessary to minimize the exclusion of participants and data loss.

5 | CONCLUSION

In conclusion, we showed that hippocampal representations for paired stimuli change dynamically over time during associative learning. In

particular, pattern separation for paired stimuli was more prominent during the early phase of learning in the posterior hippocampus, while pattern integration was more apparent during the late phase in the anterior hippocampus. These dynamic changes during learning were observed only for remembered pairs but not for forgotten pairs in subsequent memory retrieval tests. These findings suggest the existence of temporally and spatially dynamic hippocampal information processing during learning, possibly leading to the maintenance of associative memory, and provide a comprehensive view that can reconcile prior studies with regard to the role of the hippocampus during learning.

AUTHOR CONTRIBUTIONS

Minjae Kwon and Sue-Hyun Lee designed the study. Minjae Kwon collected and analyzed the data. Sang Wan Lee contributed to the data analysis. Minjae Kwon and Sue-Hyun Lee interpreted data and wrote the manuscript. Sue-Hyun Lee supervised the research.

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CONFLICT OF INTEREST

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

The behavioral and neuroimaging data are not able to be made openly available because of the need for approval from the Institutional Review Boards. Sharing and reuse of data require a reasonable request to the corresponding author and approval from the Institutional Review Boards.

ORCID

Sang Wan Lee  <https://orcid.org/0000-0001-6266-9613>

Sue-Hyun Lee  <https://orcid.org/0000-0002-5318-5657>

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